

Tracking nitrogen allocation to proteome biosynthesis in a marine microbial community

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Supplementary Materials for

**Tracking nitrogen allocation to proteome biosynthesis
in a marine microbial community**

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Supplementary Materials:

Supplementary Figure S1
Supplementary Tables S1-S2

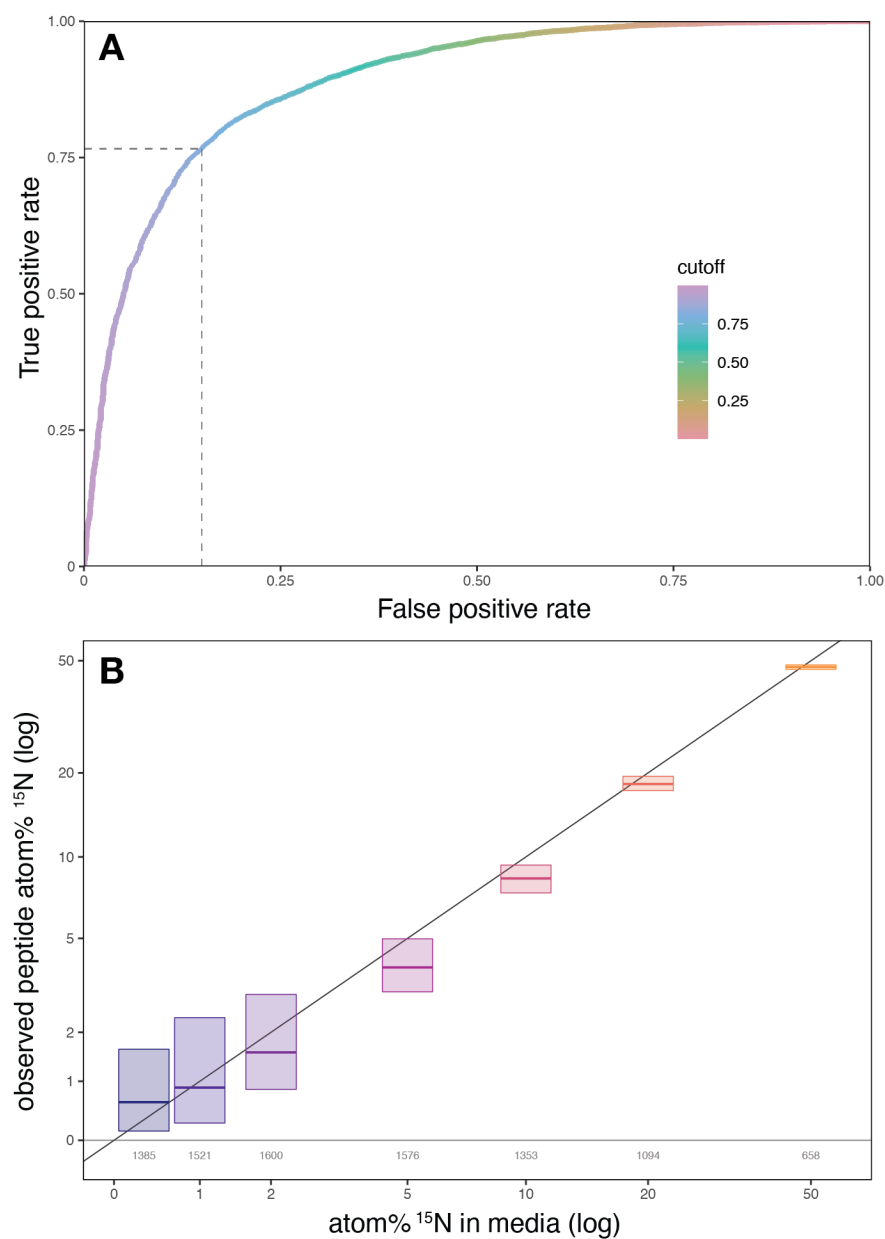


Figure S1.

Evaluation of classifier model used to filter peptide-level ^{15}N data as part of the cPIE analytical pipeline. **(A)** ROC curve for classifier model (AUC=0.89) and **(B)** observed peptide ^{15}N atom% enrichment values for peptides from *Synechococcus* WH7803 cultured on atom% ^{15}N media (median shown by cross bar with 25th and 75th percentiles as bottom and top of boxes, respectively; note log scale on both axes). WH7803 data analyzed with cPIE showed a close match between the observed ^{15}N atom% values and the expected amount of isotope incorporation.

Analysis step	# peptide datapoints	# peptide sequences (incl. variable mods)	# peptide sequences	# amendment time courses	# all time courses (incl. control)
<i>cPIE output / classifier input</i>	278735	11270	10705		
day/dusk	150171	9540	9127		
night/dawn	128564	9033	8676		
<i>Classifier output</i>	185229	8066	7827		
day/dusk	98986	6837	6656		
night/dawn	86243	6492	6344		
<i>Exclude T0 outliers</i>	185101	8056	7817		
day/dusk	98891	6829	6648		
night/dawn	86210	6486	6338		
<i>Resolve variable Met oxidation</i>	183817		7817		
day/dusk	98075		6648		
night/dawn	85742		6338		
<i>Exclude peptides w/ interferences</i>	179979		7748		
day/dusk	96082		6579		
night/dawn	83897		6269		
<i>Collapse duplicate bottles</i>	125087		7748		
day/dusk	65783		6579		
night/dawn	59304		6269		
<i>Peptides with complete timecourses</i>	53013		1672	10130	
day/dusk	30391		1561	5744	7305
night/dawn	22622		1097	4386	5483
<i>Decrease threshold</i>	48263		1605 (+67 in control only = 1672)	9062	
day/dusk	27750		1493 (+68 in control only = 1561)	5160	6721
night/dawn	20513		1057 (+40 in control only = 1097)	3902	4999
<i>Timecourse 15N incorporation rates</i>			1605	9062	
day/dusk			1493	5160	
night/dawn			1057	3902	

Table S1.

Summary of ¹⁵N-tracking proteomics dataset filtering/timecourse construction steps to arrive at the measured set of 9062 peptide ¹⁵N incorporation rates (Fig 1B). For each step, the resultant number of datapoints, sequences, or timecourses is shown for the total dataset (top row) and separately for the day/dusk and night/dawn incubations (second and third rows, respectively).

Element	Isotope abundance_Mass shift	Tracer isotope mass shift	Tracer purity
C	0.0107_1/0.9893_0	1	
O	0.99757_0/0.00038_1/0.00205_2	2	
N	0.99632_0/0.00368_1	1	0.98
S	0.9493_0/0.0076_1/0.0429_2/0.0002_4		
H	0.000115_1/0.999885_0	1	

Table S2.

IsoCorrectoR element file containing information about tracer purity and natural abundance isotopes